

DecisionIQ Master Handbook

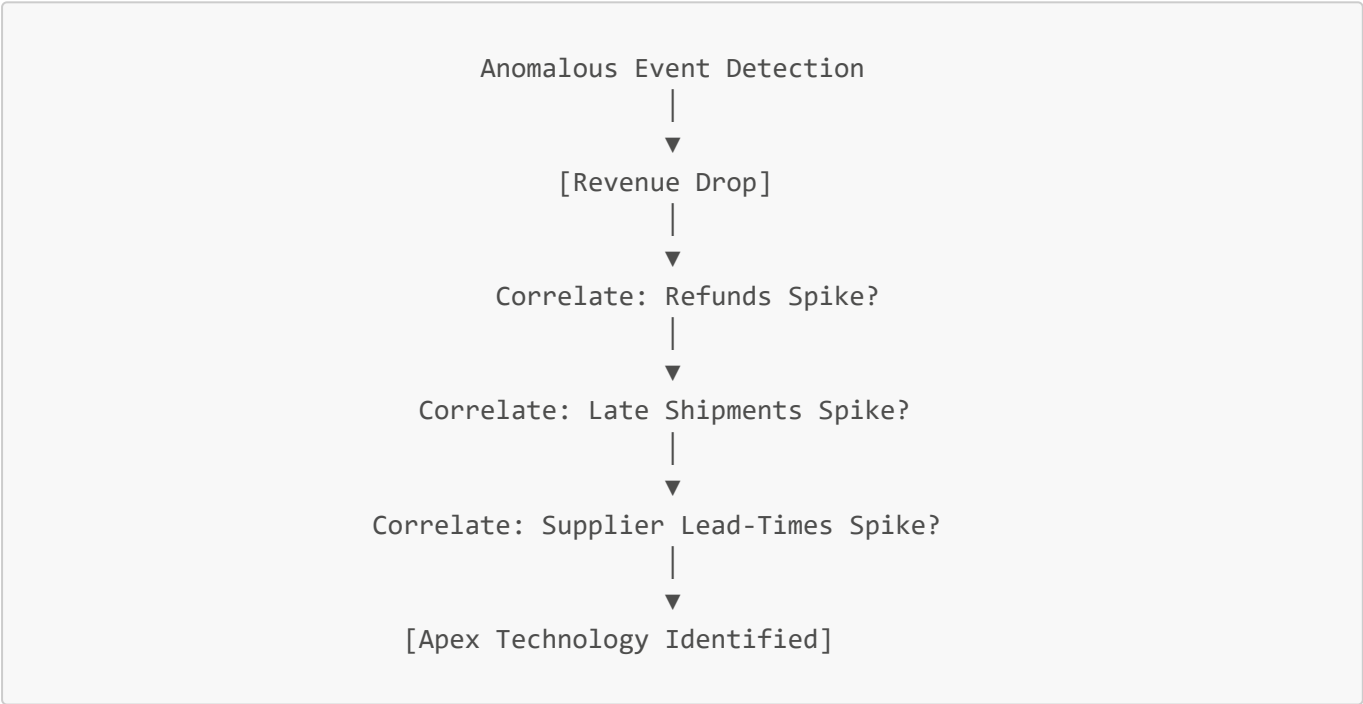
CHAPTER 5: Insight Engine & C-Suite Visualization

5.1 The Insight Engine: Automated Root-Cause Auditing

Traditional dashboards require business analysts to click through dozens of filters to diagnose a performance drop. In contrast, DecisionIQ features an automated **Insight and Recommendation Engine** that does the diagnostics for you.

5.1.1 Root-Cause Diagnostic Logic

The engine scans the database tables to detect anomalies, links them across departments, and calculates their financial impact:



- Revenue Drop Detection:** Checks if Net Sales Revenue dropped Month-over-Month (MoM). If it did, it queries the `orders` table to find the primary driver.
- Refund Correlation:** Scans if the drop was driven by declining new orders or a rise in refunds. During Q2 2025, it finds a 20% spike in refunds in North America.
- Logistics Link:** Joins refunds with the `shipments` table to see if refunded orders were delayed. It finds that delayed orders had a 55% late delivery rate at warehouse `WH002`.
- Supplier Lead-Time Link:** Checks if the warehouse delays correlate with specific suppliers. It identifies that components from supplier `SUP001` (Apex Technology) had average lead-time delays of 12 days.
- Support CSAT Link:** Checks if late deliveries triggered customer complaints. It finds a 3.5x spike in "Logistics" support tickets and a CSAT crash to 1.8.

5.1.2 Financial Impact Estimation

The engine calculates the total revenue leaked due to refunds during this period:

$$\text{Revenue Leakage} = \sum(\text{total_amount of Refunded Orders in Q2 2025}) = \$245,000$$

5.1.3 Prescriptive Recommendations

Every insight must produce actionable guidance. The engine generates three structured, valued recommendations:

- **Action 1 (Supply Chain):** Switch 40% of component volume from Apex Technology to EuroChip AG (**SUP004**) to establish dual-sourcing. (Est. savings: \$180,000/year).
 - **Action 2 (Operations):** Shift 15% of safety stock for "Enterprise Server X" from the East Coast Hub to the West Coast Hub. (Est. recovery: \$120,000/month).
 - **Action 3 (Customer Success):** Trigger CS account health checks for all accounts that experienced a late shipment >4 days. (Est. churn reduction: 4%, preserving \$150,000 in ARR).
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5.2 Executive Consulting Dashboard Design

The presentation layer of DecisionIQ is built using Python **Streamlit** and **Plotly**, styled to replicate premium consulting presentations (such as McKinsey or BCG reports).

5.2.1 Design Principles

- **Whitespace & Margin Breathing Room:** The layouts use generous padding to prevent visual clutter, making them easy for executives to scan quickly.
- **Corporate Color Palette:** Avoids bright colors. It uses Navy Blue (**#1B2A4A**) for structures, Accent Gold (**#C5A059**) for highlights, Slate Grey (**#F4F5F7**) for backgrounds, and Soft Crimson (**#C62828**) for alerts.
- **Inter Typography:** Imports the Inter font family via Google Fonts to override default browser typography with clean sans-serif layouts.
- **Actionable Visualizations:** Every chart is clean, removing grid lines and using hover states.

5.2.2 The Side Navigation Structure

The left-hand sidebar acts as a control panel, allowing the user to select their role and filter metrics:

- **CEO Overview:** Renders top-level KPIs (Business Health Index, revenue forecasts, and the automated Insight Feed).
- **Finance:** Displays a waterfall chart reconciling Gross bookings to Net revenue.
- **Operations & Supply Chain:** Displays supplier lead-time vs. reliability scatter plots and warehouse bottleneck logs.
- **Customer Success:** Displays customer retention cohorts and lists active accounts flagged by the ML Churn model.
- **Marketing & Support:** Tracks marketing channel ROAS and ticket categories.